

Classification of Tomato Ripeness Levels Using Convolutional Neural Network (CNN)

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Abstract

This research develops an image-based tomato ripeness classification system using Convolutional Neural Network (CNN). The dataset consists of 3600 tomato images classified into three classes, namely unripe, half-ripe, and ripe. These images were obtained from publicly available datasets, specifically 2,550 images from Kaggle and 1,050 images from GitHub. Before testing, a pre-processing stage is carried out which includes resizing to reduce the size of the image, cropping to focus on the main object, namely tomatoes, and augmentation to increase the generalization ability of the model. The CNN model is built with an architecture that is capable of extracting visual features automatically through the learning process. The training results for 10 epochs show that the validation accuracy reaches 99.39%, with a loss of 4.16%, indicating that the model is able to learn well without significant overfitting. These results show that CNNs perform well in accurately identifying tomato maturity, while emphasizing the possibility of further improving the model through transfer learning and optimization to handle more complex data.

Keywords: Tomato, Ripeness, Classification, CNN, Digital Image

INTRODUCTION

Tomatoes are one of the easy fruits to ripen, no need to use your hands to know when the tomatoes will ripen. Immature tomatoes are green in color, while ripe ones will turn orange and red [1]. Tomatoes have many benefits, such as being consumed in everyday life as vegetables, seasonings, table fruit, coloring materials, and cosmetics. In addition, tomatoes are also widely used in the food processing industry such as sauces, juice, and other processed products. Therefore, tomatoes are one of the horticultural commodities that are very multipurpose and have high economic value. The nutritional content is also rich, especially vitamin A and antioxidants which are beneficial for eye health, improving immunity, maintaining soft skin, and preventing hypertension and other diseases [2].

According to data from the Central Statistics Agency (BPS) in 2022, the demand and consumption of tomatoes in Indonesia have shown a consistent upward trend each year. National tomato production in 2021 was recorded at 1.12 million tons, an increase of 0.21% compared to the previous year's production of 1.11 million tons [3]. The demand for tomatoes continues to grow in domestic and international markets, requiring a consistent supply in terms of quality and quantity. In the context of exports, the demand for fresh tomatoes presents a significant opportunity. One of the important aspects in ensuring such quality is the ability to identify the ripeness level of the fruit, as ripeness directly affects the flavor, texture, nutrient content, and shelf life of the product [4]. The obvious color change exhibited by tomatoes as they ripen has attracted the attention of researchers in the field of digital image processing. Another study also stated that the visual characteristics of an object strongly support the use of deep learning technologies, such as Convolutional Neural Network (CNN), to automatically detect the ripeness of fruits [5].

Over the past few years, digital image processing has developed rapidly and is utilized in various fields, including agriculture. This technology enables automatic classification of objects, supported by advances in artificial intelligence and computer vision [6]. Digital image processing is a field of science that studies various methods of processing still and moving images digitally with the help of computers [7]. One

approach that is experiencing rapid development in this context is the Convolutional Neural Network (CNN). CNN is an effective deep learning method in recognizing visual patterns and performing image classification with a high level of accuracy, even exceeding other conventional approaches [8]. CNN is very suitable for analyzing the ripeness level of fruits, especially tomatoes, because of its ability to automatically extract visual features. This model can recognize simple features such as edges and textures to complex visual patterns, without going through manual feature engineering stages. This ability makes CNN very flexible in handling shape, color, and lighting variations in fruit images [9].

However, most of the previous studies still rely on conventional machine learning methods such as Support Vector Machine (SVM). SVM has shown good classification performance, especially when supported with manually extracted features such as color and texture [10]. However, this approach has limitations because it relies on a complex feature engineering process and is less adaptive to the dynamics of image variations. The emergence of CNN is a solution to these limitations with an end-to-end learning approach, where the model learns directly from raw data without the need for separate feature processing stages [11]. Research by Ahmad showed that CNN is able to identify tomato fruit classification accurately, even superior to conventional methods such as SVM [12].

Therefore, this research was conducted to find out how the Convolutional Neural Network (CNN) method performs in classifying the ripeness level of fruit, especially tomatoes, based on digital images. Thus, this study aims to explore the effectiveness of CNN as a modern approach in multiclass classification, as well as assess the effect of various image pre-processing techniques such as resizing, cropping, and augmentation on the accuracy of the classification model.

METHODS

In this research, a series of systematically designed stages is employed to achieve the goal of classifying tomato ripeness levels using the Convolutional Neural Network (CNN) method. The process stages are illustrated in Figure 1.



Figure 1. Flowchart of the Research Methodology

1. Data Collection

Dataset collection is the initial stage carried out by gathering information in the form of tomato images to conduct research on identifying the ripeness level of tomatoes [13]. The data were collected using a secondary method, sourced from public datasets available on Kaggle and GitHub. A total of 2,550 tomato images were obtained from Kaggle (Source: <https://www.kaggle.com/datasets/sujaykapadnis/tomato-maturity-detection-and-quality-grading>), and another 1,050 images were obtained from GitHub (Source: <https://github.com/APS360Team45/Data-Processing/tree/main/Fruits/Tomato>). A total of 3,600 tomato images were collected and categorized into three classes based on ripeness level: unripe, half-ripe, and ripe, with each class consisting of 1,200 images. More detailed information about the tomato image dataset is presented in Table 1.

2. Pre-processing

Pre-processing is the stage of preparing the tomato image for use in the model training process. The process involves several steps, namely resizing the image, cropping the image, and adding data (augmentation) [14].

3. Data Splitting

Data splitting is the stage that follows preprocessing, where the dataset is divided into two parts: 80% for training to train the model and 20% for validation to evaluate the model's performance during training,

following the Pareto distribution principle. A predictive model is considered acceptable if it can achieve a minimum sensitivity level of 80% [15].

Table 1. Tomato Image Dataset Details

Class	Number of Images	Description	Image
Unripe	1200	Tomatoes are evenly colored green	
Half-ripe	1200	The tomato has a gradient of green, yellow, and orange	
Ripe	1200	Tomatoes are evenly colored red	

4. CNN Model

CNN is a deep learning model that is specialized for computer vision so that it can identify and classify the pattern of an image. The model is inspired by the workings of the visual cortex which is designed to resemble neurons in the human brain [16].

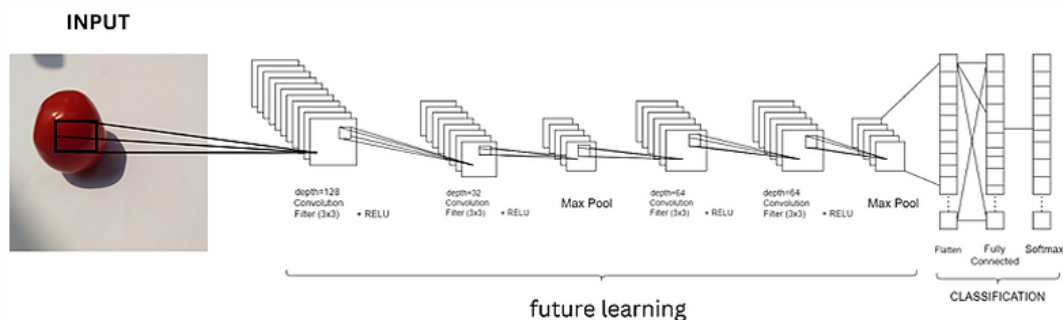


Figure 2. Convolutional Neural Network (CNN) Model

Figure 2 above illustrates the CNN model, which consists of two main stages: feature learning and classification. The feature learning stage includes a convolution layer, ReLU activation, and max pooling, while the classification stage consists of flattening, a fully connected layer, and softmax. The model uses several types of layers, including Conv2D for extracting features using various filters, MaxPooling2D for reducing the dimensionality of the data, Flatten for standardizing or flattening the data, and Dense as the final classification layer. The activation function used is ReLU to accelerate training, with a kernel size of 3x3 and a pooling size of 2x2 [13]. A summary of the CNN model used can be seen in Table 2.

Table 2. CNN Model Summary

Layer (type)	Output Shape	Param#
conv2d_3 (Conv2D)	(None, 148, 148, 32)	896
max_pooling2d_3 (MaxPooling2D)	(None, 74, 74, 32)	0
conv2d_4 (Conv2D)	(None, 72, 72, 64)	18,496
max_pooling2d_4 (MaxPooling2D)	(None, 36, 36, 64)	0
conv2d_5 (Conv2D)	(None, 34, 34, 128)	73,856
max_pooling2d_5 (MaxPooling2D)	(None, 17, 17, 128)	0
flatten_1 (Flatten)	(None, 36,992)	0
dense_2 (Dense)	(None, 128)	4,735,104
dropout_1 (Dropout)	(None, 128)	0
dense_3 (Dense)	(None, 3)	387

Total params: 4,828,739

Trainable params: 4,828,739

Non-trainable params: 0

The CNN architecture consists of 3 convolution layers with 3x3 filters of 32, 64, and 128. Each layer is followed by 2x2 max pooling to reduce the feature dimension, after which the results are flattened by flatten and then processed by a dense layer with 128 neurons and ReLU as the activation function, and to prevent overfitting a dropout of 0.5 is used. The output layer uses softmax with 3 neurons according to the number of classes. The model shows a total of 4,828,739 parameters can be trained.

5. Testing

Testing is a stage that is carried out after passing pre-processing data, making test data, compiling training data and designing models. This data testing aims to ensure that the Convolutional Neural Network (CNN) model is able to identify the level of tomato maturity properly [17]. In this test, there are two main indicators measured, namely accuracy and loss. Accuracy shows the percentage of correct predictions from the total test data and reflects how well the model can classify correctly. Meanwhile, loss measures the level of difference between predictions and actual values; a low loss value indicates that the model is able to learn well and produce accurate predictions [18]. The best CNN model is selected based on training accuracy, training loss, validation accuracy, and validation loss. This ensures that the resulting model not only performs well on the training data but also generalizes well to the test data [19].

RESULT AND DISCUSSION

1. Pre-processing

a) Resize

The original tomato image is large, measuring 6000 x 8000 pixels, and is then resized to 250 x 250 pixels to make it more uniform and efficient for processing in the CNN model. This resizing process was performed using the Keras library and the Python Imaging Library (PIL), which employ interpolation techniques to maintain the visual quality of the images. By using a consistent size, the CNN can extract features without requiring extensive computational resources.

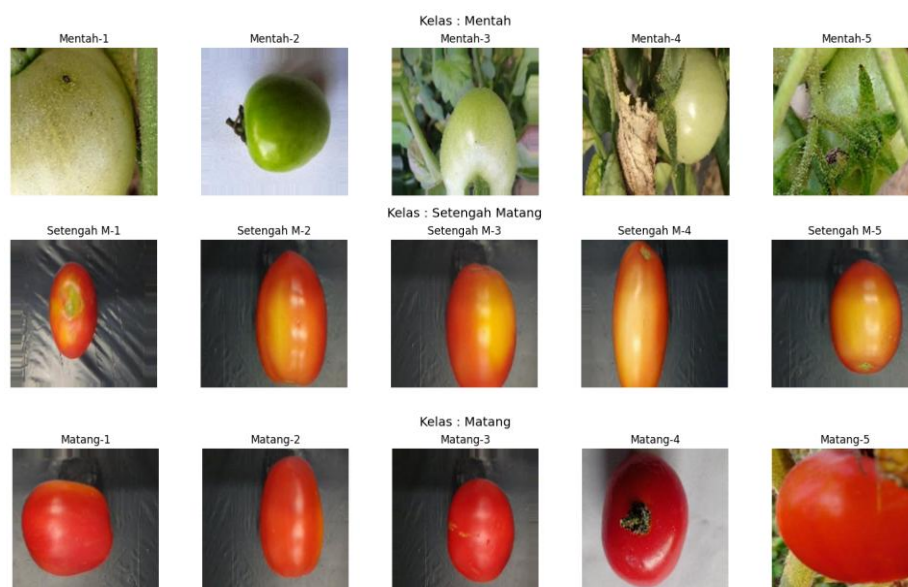


Figure 3. Tomato Pre-processing Result

b) Cropping

After the image was resized to 250 x 250 pixels, the center portion was cropped to 150 x 150 pixels for this study. By removing the unimportant edges, this cropping focuses on the tomato image. To perform this

process, the central area of the image is taken symmetrically. This ensures that the main features of the tomato remain intact and become more prominent during the CNN model training process.

c) Augmentation

Initially, 3,600 images were included in three classes. The dataset grew to 9,102 images after augmentation using the Keras module with techniques such as pixel normalization, angle distortion (shear), random zoom, and horizontal flipping. The purpose of this augmentation was to enhance the data and improve the model's ability to identify tomatoes in a variety of visual conditions. The results of these three pre-processing stages are shown in Figure 3.

2. CNN Model

The CNN model used proved to be very effective in classifying the ripeness level of tomatoes. With 10 epochs, the training accuracy increased from 88.81% to 99.32% with a final training loss of 0.0269. The highest validation accuracy was achieved at epoch 6 with 99.61%, accompanied by the lowest validation loss of 0.0251, indicating good generalization to new data. Within a time range of 386–439 seconds per epoch, the model was able to learn to recognize tomato visual features with high accuracy and minimal error. The results are summarized in Table 3, which outlines the performance metrics of the model trained with validation.

Table 3. Results of CNN model training for 10 epochs

Epoch	Accuracy	Loss	Validation Accuracy	Validation Loss	Waktu (detik)
1	0.8881	0.2871	0.9730	0.0879	429
2	0.9820	0.0610	0.9901	0.0630	401
3	0.9877	0.0489	0.9757	0.0798	394
4	0.9895	0.0379	0.9658	0.1289	387
5	0.9866	0.0568	0.9928	0.0357	439
6	0.9936	0.0250	0.9961	0.0251	388
7	0.9952	0.0230	0.9934	0.0450	395
8	0.9935	0.0232	0.9840	0.0582	386
9	0.9852	0.0564	0.9923	0.0501	395
10	0.9932	0.0269	0.9939	0.0416	407

3. Testing

Figure 4 shows that the CNN model successfully extracted key visual features from the tomato image, including color, texture, and shape to determine the ripeness level. From the three samples tested, the model showed high accuracy in classifying tomatoes into unripe, half-ripe, and ripe categories with a good match between predicted and actual labels.

This classification success demonstrates the CNN's ability in optimal feature learning, supported by data augmentation that increases the model's robustness to variations in lighting, viewing angle, and background. The results confirm that the CNN model has good classification performance for image-based tomato ripeness detection.



Figure 4. Tomato Image Prediction Testing

Based on previously conducted experiments, this presentation includes evaluation metrics such as accuracy and loss on training data and validation data, to provide a clear picture of the performance and effectiveness of the CNN model in performing classification can be seen in Figure 5 and Figure 6.

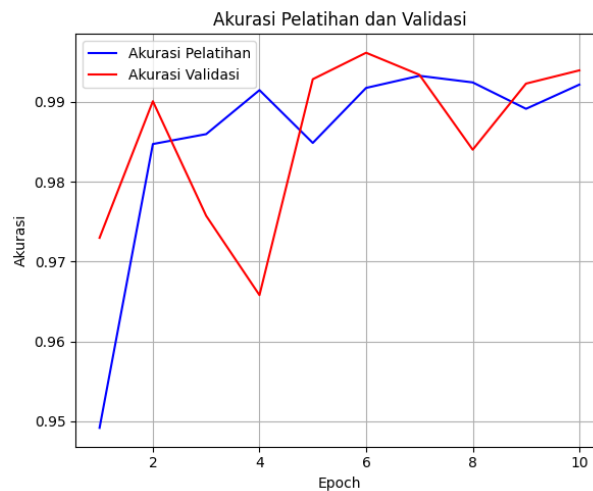


Figure 5. CNN Model Training and Validation Accuracy Graph

Figure 5 shows the accuracy of the training (blue line) and validation (red line) data during the CNN model testing process. It can be seen that the increase in training accuracy is stable from about 95% at the first epoch to 99.22% at the tenth epoch. The accuracy on the validation data is also quite good although there are slight ups and downs, with the highest value of almost 99.39% at the 6th epoch. These results show that the model can learn well and is able to recognize new data well too.

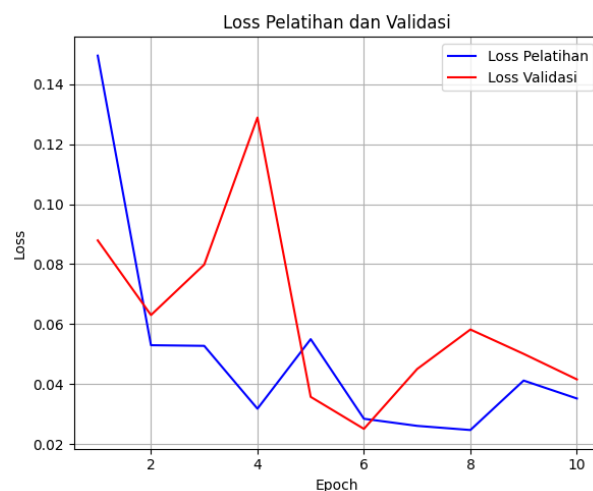


Figure 6. CNN Model Training and Validation Loss Graph

Figure 6 shows the accuracy of the training (blue line) and validation (red line) data during the CNN model testing process. It can be seen that the loss value on the training data decreases gradually from a high value at the beginning until it reaches about 3.52% at the tenth epoch. Meanwhile, the loss on the validation data shows fluctuations with some increases and decreases, but finally drops to a value of about 4.16% at the end of training.

Fluctuations in validation loss and accuracy during the training process are a common scenario in deep learning model training. This phenomenon reflects unstable convergence dynamics and can be caused by factors such as overfitting or data imbalance. This aligns with the findings of research conducted by Saleem, Potgieter, and Arif, who observed the same fluctuation patterns while training deep learning models for plant disease classification. Therefore, this represents a unique development in deep learning models and is analyzed to ensure the model's accuracy and stability are optimized [20].

CONCLUSION

Based on the results of the research conducted, it can be concluded that the Convolutional Neural Network (CNN) model is effective in identifying the ripeness level of tomatoes. The developed model consists of three main convolutional layers and utilizes preprocessing techniques such as resizing, cropping, and augmentation, which have proven to enhance the model's ability to generalize to new data.

From the testing results over 10 epochs using tomato images classified into three categories unripe, half-ripe, and ripe the model achieved a validation accuracy of 99.39% and a loss value of 4.16%. This indicates that the model was able to learn effectively without experiencing significant overfitting.

This research contributes valuable insights into the study of digital image processing based on deep learning, particularly in the context of precision agriculture. In the future, the model can be further developed by incorporating optimization techniques or transfer learning to improve its efficiency and adaptability to data with more complex backgrounds or lighting conditions.

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